Lab 01: Windows Process-Event Triage

A safe beginner exercise using sample evidence and optional local Event Viewer inspection

Learning goals

- Distinguish an event from an alert and an incident.
- Identify key fields in a process-creation record.
- Write an evidence-based triage note without overclaiming.

SAFETY / COST Use the included sample data first; it costs nothing and changes no system settings. Only inspect logs on a device/account you are authorised to use.

Part A — Read the sample record

Open the companion file website/public/labs/windows-process-creation-samples.json, or use the sample-data download button on this lab card. It contains intentionally simplified training records with synthetic IDs. Do not treat them as live telemetry.

Field	Sample value	Interpretation
event_id	SAMPLE-4688-001	Training identifier, not a real Windows ID
windows_event_id	4688	Windows Security process-creation event ID, if auditing is configured
host	TRAINING-WIN-01	Synthetic host label
process	C:\Windows\System32\notepad.exe	Harmless example process path
user	STUDENT-LAB	Synthetic username
label	benign	Expected label for this training record

Part B — Optional local verification

1. Open Event Viewer on your own Windows computer (search the Start menu for Event Viewer).

2. Browse to Windows Logs > Security. Access may require appropriate permissions; do not change policy if you are unsure.
3. If process-creation auditing is already enabled, inspect a recently created benign process such as Notepad and locate the relevant event record.
4. If no matching event appears, use the bundled sample data. Process-creation auditing may not be enabled by default; this exercise does not require changing group policy.

Part C — Investigate the training case

The second sample is marked suspicious for the purpose of exercising a rule because its command-line field contains an encoded-command indicator. This label is a training ground truth, not proof that a real user or device is compromised.

1. Record the event type, host, user, process, and command-line context.
2. Describe the exact condition that caused the rule to match.
3. List at least two pieces of additional context you would seek in a real investigation.
4. Describe one legitimate reason administrative PowerShell may be used.
5. Write a short conclusion using “observed”, “suggests”, and “not yet confirmed” accurately.

Answer key / discussion

The benign Notepad record should not match a rule scoped to PowerShell encoded-command indicators. The labelled PowerShell training record should match if the rule checks the process name and relevant argument marker. That rule is only a triage signal; analyst review is needed to establish intent and scope.

SUCCESS CRITERIA Your note cites observable fields, explains the rule match, identifies missing evidence, and avoids claiming that a rule match alone proves malicious intent.

Further study

- Windows event auditing overview (Microsoft Learn)
- Wazuh getting started

A real lab for these notes

Topic: Windows Security events, process creation, and safe triage. These pages connect the existing OpenCyberPath notes to specific provider-hosted exercises. The card label is intentionally generic; the exact task name and direct URL are listed here so you can open the right exercise instead of a whole catalogue.

Concept checkpoints

- Event ID 4688 describes process creation when the relevant audit policy is configured.
- Read the process path, user, command line, parent process (if available), host, and timestamp as a group.
- A command-line indicator is a lead for review, not proof of maliciousness.
- Use synthetic sample data first; live event-log access may need permission.

Windows Event Logs *[TryHackMe]*
https://tryhackme.com/room/windowseventlogs
Access note: Account required; the provider page currently marks substantial tasks as paid-tier. Check access before relying on it. A deployable lab machine is described on the page.

Windows Logging for SOC *[TryHackMe]*
https://tryhackme.com/room/windowsloggingforsoc
Access note: Account required; provider entitlements can change. The room focuses on Windows logs, Sysmon, process events, and authentication examples.

ACCESS CHECK Open the exact task page and verify the current login, account tier, VM launch button, free/paid terms, and any cost notes before starting. A provider may change availability or retire a challenge.

Before you click Start

- Write down the provider's target/scope and use only the instance it creates for you.
- Keep this notebook beside the exercise. Use its original course data before trying commands on a local machine.
- Capture timestamps, filenames, row/packet IDs, and a short explanation; redact credentials, tokens, and personal data.
- If a page says retired, premium, or downloadable-only, use the alternative listed in the access note or return to the matching interactive lab above.

Commands you can explain

Only inspect logs on your own device or an explicitly assigned lab VM. These commands are read-only and do not change audit policy.

First download the matching sample file(s) from the existing card when a command names one. Keep the data in a practice folder and never execute strings stored in evidence.

POWERSHELL / SAMPLE JSON

```
Get-Content .\windows-process-creation-samples.json -Raw | ConvertFrom-Json | Format-Table
event_id,windows_event_id,host,user,process,activity_label -AutoSize
```

Read the bundled training fixture. Download the sample JSON beside the command before running it.

POWERSHELL / LIVE LOCAL LOG

```
Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4688} -MaxEvents 20 | Select-Object
TimeCreated,Id,ProviderName,Message
```

Read-only query of the current computer's Security log. It can return access denied or no events if auditing is not configured; use the sample JSON instead.

POWERSHELL / FAILED LOGONS

```
Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4625} -MaxEvents 20 | Select-Object
TimeCreated,Id,Message
```

Review a small number of failed-logon events on a device you are authorised to inspect.

COMMAND PROMPT

```
wevtutil qe Security /q:"*[System[(EventID=4688)]]" /c:10 /f:text
```

Read up to ten process-creation records on an authorised Windows machine. Access and audit configuration may limit output.

POWERSHELL / VERIFY A DOWNLOADED SAMPLE

```
Get-FileHash .\windows-process-creation-samples.json -Algorithm SHA256
```

Record a file hash when preserving the exact sample used for the exercise.

COMMAND HABIT Check the current working directory, shell, target, and expected effect before pressing Enter. Save the exact command and the output that mattered.

Guided lab run

- 1. Open the direct Windows logging room and start its attached machine if available.
- 2. Start with the included sample JSON if the provider room is unavailable.
- 3. For each record, note timestamp, host, user, process, command line, parent process (if present), and training label.
- 4. Compare the harmless Notepad example with the explicitly redacted placeholder example. Do not execute the placeholder command.
- 5. Write one observed fact, one plausible interpretation, one alternative explanation, and one next check.

Evidence checklist

- [ ] What exact lab instance or sample file was used?
- [ ] Which command/query and time range produced the observation?
- [ ] What is the directly observed fact, separate from your inference?
- [ ] What benign/alternative explanation could fit?
- [ ] What extra record or control would help confirm the conclusion?

Check your understanding

Q *Does Event ID 4688 prove a program is malicious?*

Think it through: No. It records process creation. The command, parent, user, timing, signature, and surrounding events provide context.

Q *Why are placeholders marked REDACTED?*

Think it through: They are intentionally non-executable training markers and should never be treated as runnable commands.

Q *What should you do if Get-WinEvent returns access denied?*

Think it through: Use the bundled sample fixture or the provider's lab VM; do not change security settings or elevate on a managed device.

NOTEBOOK CLOSE-OUT Finish with: Observation / Evidence source / Why it matters / Alternative explanation / Next check / Limitation. A careful "not enough evidence yet" is a valid professional result.